

Biology

The science that focuses on the study of living things is broadly described as biology. The people who study biology are called biologists, and they examine every form of life—from the workings of

microscopic organisms, such as green algae, to the way large groups of animals behave. They also investigate how living organisms survive, where they originated, and how they interact with each other.

What does it consist of?

Biology covers a number of different fields. The two main areas are zoology and botany, but there are many other branches that focus on specific areas of study.



Zoology

The study of animals is called zoology. Zoologists look at the structure of animals and examine how they live and behave.



Botany

The study of plants is called botany. Botanists investigate all kinds of plant life, from mosses and ferns to trees and cacti.



Microbiology

The study of microorganisms is called microbiology. This includes cells that are too tiny to be seen without a microscope.



Ecology

The branch of biology that examines the relationship between living things and their environment is called ecology.



Medicine

The science of the prevention, diagnosis, and treatment of disease is called medicine.



Paleontology

Paleontologists study the fossilized remains of plants and animals in order to understand how different species lived and evolved.

Requirements for life

Living things have certain life processes in common. They all share the same seven characteristics that set them apart from nonliving things.



Reproduction

All living things reproduce and create new generations of a species (offspring). The process of reproduction enables a species to survive and spread.



Respiration

Chemical and physical processes where an intake of oxygen is conveyed to tissues and cells, and carbon dioxide is exhaled.



Sensitivity

Most living things use their senses to detect changes in their surroundings, such as light levels or changes in the weather. Animals depend on rapid responses for survival.



Excretion

Excretion is the process of getting rid of waste. Humans, for example, excrete urine and carbon dioxide (the gas produced during respiration).



Movement

All living things can move in some way—plants can move their leaves and flowers to respond to light, and animals move to find food, to mate, or to escape predators.



Growth

Most living things grow by making extra cells. As they grow, they also develop, taking on new shapes or features that enable them to function in new ways.



Nutrition

Animals need nutrients to provide the energy to make cells work. Not all living things get their energy from nutrients. Plants get their energy direct from sunlight.